Yakeen Plus (2025)

SHORT PRACTICE TEST - 06

DURATION : 60 Minutes | DATE : 20/10/2024 | M. MARKS : 192

ANSWER KEY

PHYSICS

1.	(3)
2.	(3)
3.	(1)
4.	(3)
5.	(1)
6.	(4)
7.	(4)
8.	(4)
9.	(2)
10.	(3)
11.	(2)
12.	(3)

CHEMISTRY

13.	(4)
14.	(1)
15.	(2)
16.	(1)
17.	(1)
18.	(4)
19.	(2)
20.	(2)
21.	(2)
22.	(2)
23.	(3)
24.	(1)

BOTANY

25.	(2)
26.	(1)
27.	(2)
28.	(2)
29.	(3)
30.	(2)
31.	(4)
32.	(2)
33.	(1)
34.	(1)
35.	(4)
36.	(4)

ZOOLOGY

37.	(3)
38.	(3)
39.	(4)
40.	(1)
41.	(2)
42.	(2)
43.	(3)
44.	(3)
45.	(2)
46.	(1)
47.	(4)
48.	(3)

SECTION-(I) PHYSICS

1. **(3)**

Path difference $\Delta x = \frac{\lambda}{2\pi} \times \varphi \Rightarrow 1 = \frac{\lambda}{2\pi} \times \frac{\pi}{2} \Rightarrow \lambda = 4m$

Hence $v = n\lambda = 120 \times 4 = 480$ m/s

[New NCERT Class 11th Page No. 371]

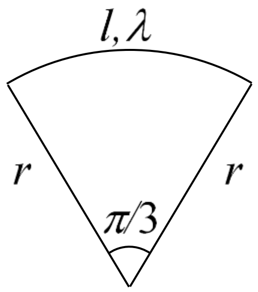
2. **(3)**

Length of the arc $= r\theta = \frac{r\pi}{3}$

Charge on the arc $= \frac{r\pi}{3} \times \lambda$

$\therefore$ Potential at center $= \frac{kq}{r}$

$= \frac{1}{4\pi\varepsilon_0} \times \frac{r\pi}{3} \frac{\lambda}{r} = \frac{\lambda}{12\varepsilon_0}$



3. **(1)**

Metal plate acts as an equipotential surface, therefore the field lines should enter normal to the surface of the metal plate.

[New NCERT Class 12th Page No. 068]

4. **(3)**

$v \propto \sqrt{T} \Rightarrow \sqrt{\frac{T_2}{T_1}} = \frac{v_2}{v_1} \Rightarrow T_2 = T_1 \left(\frac{v_2}{v_1}\right)^2$

$\Rightarrow T_2 = 273 \times 4 = 1092K$

[New NCERT Class 11th Page No. 376]

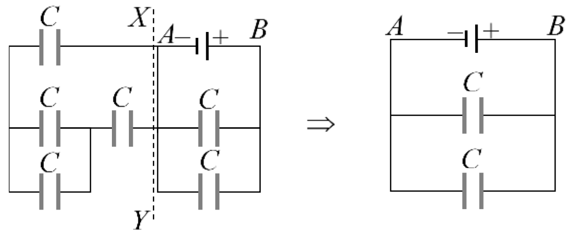
5. **(1)**

The amplitude is a maximum displacement from the mean position.

[New NCERT Class 11th Page No. 344]

6. **(4)**

All capacitor lying in left side of line XY are short circuited so circuit can be reduced as follows



$C_{AB} = 2C$

[New NCERT Class 12th Page No. 078]

7. **(4)**

Since charge flows from high potential to lower potential.

If positive charge is given, then $V_1 < V_2$ as $r_1 > r_2$

So positive charge flows from $Q \rightarrow P$

If negative charge is given, then $V_1 > V_2$

So negative charge flows form $P \rightarrow Q$.

Since it is not given that whether the charge given is positive or negative, hence the information is incomplete.

8. **(4)**

Comparing given equation with standard equation of progressive wave. The velocity of wave

$$v = \frac{\omega(\text{Co-efficient of } t)}{k(\text{Co-efficient of } x)} = \frac{200\pi}{0.5\pi} = 400\,\text{cm/s}$$

[New NCERT Class 11th Page No. 373]

9. **(2)**

$U = \frac{1}{2}C_{eq}V^2 = \frac{1}{2}(nC)V^2$

[New NCERT Class 12th Page No. 081]

10. **(3)**

$E = \frac{V}{d} \Rightarrow \frac{\sigma}{2\varepsilon_0} = \frac{V}{d} \Rightarrow d = \frac{V \times 2\varepsilon_0}{\sigma} = \frac{50 \times 2 \times 8.85 \times 10^{-12}}{0.1 \times 10^{-6}}$

$= 8.85 \times 10^{-3}\ m = 8.85\ mm$

[New NCERT Class 12th Page No. 061]

11. **(2)**

conceptual Que

[New NCERT Class 11th Page No. 348]

12. **(3)**

Acceleration $A = \omega^2 y \Rightarrow \omega = \sqrt{\frac{A}{y}} = \sqrt{\frac{0.5}{0.02}} = 5$

Maximum velocity $v_{max} = a\omega = 0.1 \times 5 = 0.5$

[New NCERT Class 11th Page No. 348]

SECTION-(II) CHEMISTRY

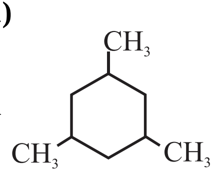
13. (4)

$$\begin{array}{c} \quad\quad CH_3 \quad CH_3 \\ \quad\quad | \quad\quad\; | \\ CH_3-CH-CH-CH_3 \end{array}$$

2, 3 dimethyl butane has 12 primary and 2 tertiary hydrogen atom

[New NCERT Class 11th Page No. 258]

14. (1)

In 

1° carbon $= 3$

2° carbon $= 3$

3° carbon $= 3$

[New NCERT Class 11th Page No. 263]

15. (2)

Order of –I effect is $-\overset{\oplus}{H_3N} > -COOH > -F > -OCH_3$

$\overset{\oplus}{H_3N}-CH_2-COOH$ has strongest acid due to strong – I Effect of $\overset{\oplus}{H_3N}$. As – I effect increases, dissociation of COOH (COO^- & H^+) increases, hence acidic nature increases.

[New NCERT Class 11th Page No. 274]

16. (1)

In hyperconjugation $\sigma \rightarrow p$ (empty) electron delocalization for tert-butyl carbocation and $\sigma \rightarrow \pi^*$ electron delocalization for 2-butene will take place.

[New NCERT Class 11th Page No. 277]

17. (1)

One double bond has one degree of unsaturation.

One triple bond has two degree of unsaturation and one ring has one degree of unsaturation.

So degree of unsaturation = 2 + 2 × 2 + 2 = 8

18. (4)

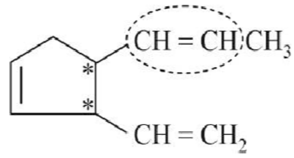
$$\overset{1}{C}H_3\overset{2}{C}H_2\overset{3}{C}H(CH_2CH_3)-\overset{4}{C}H_2-\overset{5}{C}H(CH_3)-\overset{6}{C}H_2\overset{7}{C}H_3$$ (Correct numbering)

Since, substituents are in equivalent position so lower number is given to ethyl which comes first in the name according to alphabetical order.

So, the correct name is 3-Ethyl-5-methylheptane.

[New NCERT Class 11th Page No. 263]

19. (2)



Number of stereoisomers $= 2^3 = 8$

[New NCERT Class 11th Page No. 277]

20. (2)

Lewis acids are **electrophiles.** Electrophiles love electrons, or negative charge. Boron has an empty 2p orbital and there exists a strong partial positive charge on the boron due to the extremely electronegative fluorine atoms covalently bound to boron. This strong partial positive character, coupled with a vacant orbital, makes BF_3 a potent Lewis acid and thus an electrophile.

[New NCERT Class 11th Page No. 273]

21. (2)

$$C_2H_5COONa + NaOH \xrightarrow[\Delta]{CaO} C_2H_6 + Na_2CO_3$$

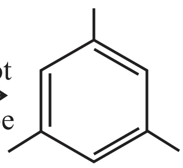
[New NCERT Class 11th Page No. 379]

22. (2)

Molybdenum oxide converts alkane to aldehyde i.e propane to propanal.

[New NCERT Class 11th Page No. 382]

23. (3)

$3CH_3-C\equiv CH \xrightarrow[\text{Cu tube}]{\text{Red Hot}}$ 

[New NCERT Class 11th Page No. 396]

24. (1)

Geometrical isomerism is not possible at site - I

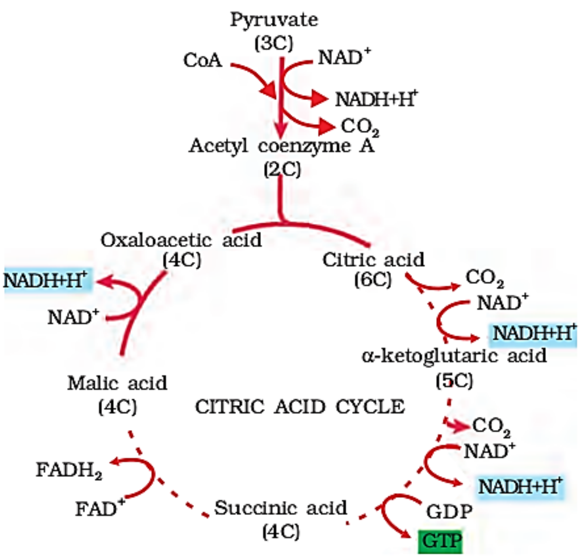
[New NCERT Class 11th Page No. 271]

SECTION-(III) BOTANY

25. **(2)**
The anatomy of the monocotyledonous root is similar to the dicotyledonous root in many respects. As compared to the dicot root which have fewer xylem bundles, there are usually more than six (polyarch) xylem bundles in the monocot root. Pith is large and well developed.
[New NCERT Class 11th Page No. 74]

26. **(1)**
In fermentation there is a net gain of only two molecules of ATP for each molecule of glucose degraded to pyruvic acid.
[New NCERT Class 11th Page No. 162]

27. **(2)**



[Old NCERT Class 11th Page No. 159]

28. **(2)**
- Sclerenchyma consists of long, narrow cells with thick and lignified cell walls having a few or numerous pits.
- The fibres are thick-walled, elongated and pointed cells, generally occuring in groups, in various parts of the plant.
- The collenchyma occurs in layers below the epidermis in dicotyledonous plants. It is found either as a homogeneous layer or in patches. It consists of cells which are much thickened at the corners due to a deposition of cellulose, hemicellulose and pectin.

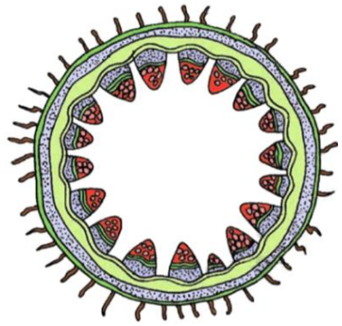
[Old NCERT Class 11th Page No. 86]

29. **(3)**
Ubiquinone receives reducing equivalents via $FADH_2$ and NADH.
[New NCERT Class 11th Page No. 160]

30. **(2)**
The tangential as well as radial walls of the endodermal cells have a deposition of water-impermeable, waxy material suberin in the form of casparian strips.
[New NCERT Class 11th Page No. 74]

31. **(4)**
The site of glycolytic pathway in case of organism undergoing aerobic and anaerobic respiration in cytoplasm.
[New NCERT Class 11th Page No. 156]

32. **(2)**



The following diagram is a transverse section of dicot stem.
[New NCERT Class 11th Page No. 75]

33. **(1)**
It would be better to consider the respiratory pathway as an amphibolic pathway rather than as a catabolic one.
[New NCERT Class 11th Page No. 164]

34. **(1)**
(1) Glucose $\rightarrow$ 1
(2) Protein $\rightarrow$ 0.9
(3) Tripalmitin $\rightarrow$ 0.7
(4) Fats $\rightarrow$ Less than 1
[New NCERT Class 11th Page No. 164]

35. **(4)**
- In grasses, certain adaxial epidermal cells along the veins modify themselves into large and empty cells. These are called bulliform cells.
- Bulliform cells in the leaves make the leaf surface exposed.
- Bulliform cells in the leaves make the leaves curl inwards.
- Isobilateral leaf shows parallel venation.

[New NCERT Class 11th Page No. 77]

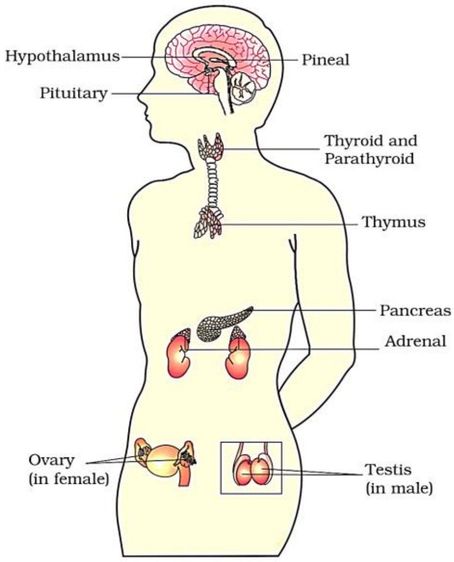
36. **(4)**
The stomatal aperture, guard cells and the surrounding subsidiary cells are together called stomatal apparatus.
[New NCERT Class 11th Page No. 72]

SECTION-(IV) ZOOLOGY

37. (3)
- The unmyelinated nerve fibre is enclosed by a Schwann cell that does not form a myelin sheath around the axon.
- They are commonly found in autonomous and somatic neural systems.

[New NCERT Class 11th Page No. 232]

38. (3)



- Melanocytes stimulating hormone is released from the pars intermedia of the pituitary gland.
- In the given diagram 'III' represents the pituitary gland.

[New NCERT Class 11th Page No. 240, 241]

39. (4)
Immediately after the influx of Na^+ ions during the generation of an action potential, the membrane at the stimulated site becomes depolarised.

[New NCERT Class 11th Page No. 233]

40. (1)
- Hypothalamic hormones reach the pituitary gland through a portal circulatory system and regulate the functions of the anterior pituitary.
- The pituitary gland is located in a bony cavity called sella tursica.
- Impairment affecting synthesis of antidiuretic hormone (ADH), might result in diabetes insipidus.
- Thyrocalcitonin (TCT), regulates the blood calcium levels.
- The process of bone resorption (dissolution/ demineralisation) is stimulated by parathyroid hormone.

[New NCERT Class 11th Page No. 240, 241, 243]

41. (2)
- Cortisol stimulates the RBC production.
- Cortisol is the main glucocorticoid in our body.
- Cortisol produces anti-inflammatory reactions.
- Cortisol is involved in carbohydrate metabolism.

[New NCERT Class 11th Page No. 245]

42. (2)
- In the inner part of the cerebral cortex, the fibres of the tracts are covered with myelin sheath and give an opaque white appearance to the layer and, hence, is called the white matter.
- The association areas in the cerebral cortex are responsible for complex functions like intersensory associations, memory and communication.

[New NCERT Class 11th Page No. 236]

43. (3)

Amino acid derivative	Epinephrine
Peptide hormone	Glucagon
Steroid hormone	Progesterone
Iodothyronine	Tetraiodothyronine

[New NCERT Class 11th Page No. 248]

44. (3)
- The hypothalamus contains a number of centres that control body temperature, urge for eating and drinking.

[New NCERT Class 11th Page No. 236]

45. (2)
- Three major regions make up the brainstem; midbrain, pons and medulla oblongata.
- Brain stem forms the connections between the brain and spinal cord.

[New NCERT Class 11th Page No. 236]

46. (1)
- Estrogen is secreted by the gonads.
- Somatostatin (growth hormone inhibiting hormone) is secreted by the hypothalamus.
- Melatonin is secreted by pineal gland.
- Catecholamines are secreted by Adrenal glands.

[New NCERT Class 11th Page No. 240, 242, 244, 246]

47. **(4)**

The primary advantage of impulse transmission at electrical synapses is that they transmit impulses faster due to direct current flow.

[New NCERT Class 11^{th} Page No. 234]

48. **(3)**

Glucagon is a hyperglycemic hormone as it increases blood glucose levels by reducing the cellular glucose uptake and utilisation.

[New NCERT Class 11^{th} Page No. 245]